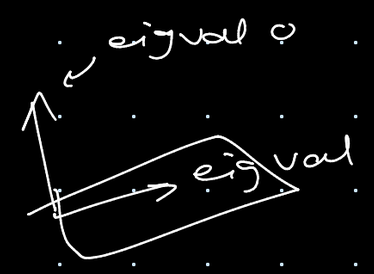
॥श्री गणेशायनमः॥

Extra points (missed in SN)

Regarding projections:

- For $A(A^TA)^{-1}A^T$ to exist, A has LI cols
- Eigen val of projection matrix P is $0/1$ → eigen vec is in same space
  - eigenvec is $\perp$ to space
- Projec^n matrix P is always diagonalizable
- P_1 projects into Ω; P_2 projects into $\Omega^\perp$. Then $P_1 + P_2 = I$, $P_1P_2 = 0$



Regarding subspaces:

- For 2 ortho subspaces (say the col^m space and the left null space) they intersect at the $\vec{0}$ only.

  If $\vec{v} \in col^m(A)$ & $\vec{v} \in N(A^T)$, then $\vec{v} = \vec{0}$.

Regarding eigenvalues

- To check singularity, check if one of the eigenvalues is 0
- Eigvals corresp. to $\lambda = 0 \in$ Null space
- $\lambda \neq 0 \in$ col^m space
- If sum of rows is same, that is an eig value. Corresp. eig. vector is $\begin{bmatrix}1\\1\\1\end{bmatrix}$
- Think of λ when checking non zero eigen values of some A^2.
- If any $\lambda < 1$, then as $n \to \infty$, $\|A^n x\|$ will tend to 0.
  If all λ's are ≥ 1, then it will tend to ∞.

Regarding types of matrices

$$A = \underbrace{\frac{A+A^T}{2}}_{\text{symm.}} + \underbrace{\frac{A-A^T}{2}}_{\substack{\text{skew}\\ \text{symm.}}}$$

Regarding Diagonalization

- If $A^2 = A \longrightarrow A$ is diagonalizable

Regarding AB:

- If $AB = BA \ \forall B$ then $A = \alpha I$ (for some scalar α)

Regarding A^TA:

- Trace of $A^TA = \sum a_{ij}^2$ where a_{ij} are elements of A.
- For every rank 1 matrix, singular value $\to \sigma$ (only 1)

$$\sigma^2 = \sum a_{ij}^2$$

Wrong Ones

(LA-12, Q8) (1:05)

Q 8

The nullspace $N(A)$ of the real (square) matrix A is spanned by the vector $v = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$.

→ In $N(A^TA)$ as well. one eig. val = 0

(a) Find an eigenvector and corresponding eigenvalue of the matrix:

$$B = (3I - A^\top A)(3I + A^\top A)^{-1}.$$

} func^n of A^TA.
∴ eigen val. → f(λ)
eigen vec → same

(Give just one eigenvalue with its eigenvector.)

(b) Aside from the eigenvalue identified in part (a), all other eigenvalues λ of B must be:

- Purely real ✓
- Purely imaginary
- Zero
- Negative real part
- Positive real part
- $|\lambda| < 1$ ← those other than 0.
- $|\lambda| > 1$
- $|\lambda| \leq 1$
- $|\lambda| \geq 1$.

$B = (3I - A^TA)(3I + A^TA)^{-1}$

one eig. val. of $A^TA = 0$ as Null space is not empty ∴ cols are dep. Det = 0

for B → $\frac{3-0}{3+0} = 1$ } one eig. val. of B is 1

Also, they'll have the same eigen vec. (B is func^n of A^TA)

corresp. to eig. val 0 → eig. vec. is null space.

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Null space = 2
2 eig. vecs corresp. to $\lambda = 0$
$\lambda = 0, 0$] → atleast 2 zero eig. vals.

$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ rank = 1, nullity = 2 } $\lambda = 0, 0$

trace = $\lambda_1 + \lambda_2 + \lambda_3 = 3$
$0 + 0 + \lambda_3 = 3$
$\lambda_3 = 3$

trace (A) = Sum of eig val (A)

$B = (3I - A^TA)(3I + A^TA)^{-1}$

↳ Dim(N(A)) = 1

rank = # of non-zero eig. values

3 non-zero eig. values.

For λ of A

$3 \to \frac{3-\lambda}{3+\lambda}$ $\quad \lambda > 0$

$\therefore \frac{3-\lambda}{3+\lambda} < 1$

Regarding SVD:

Σ essentially stretches to ellipse.

singular values decide length change.

orthogonal matrices do not change length.

NOTE: $Q_1 A Q_2$ has same σs as A.

For hermitian matrices (and symm) $\sigma_i = |\lambda_i|$

- max value of $\frac{\|Ax\|}{\|x\|}$ equals σ_1
- $\sigma_n \|x\| \leq \|Ax\| \leq \sigma_1 \|x\|$

$\sigma_1 \rightarrow$ biggest
$\sigma_n \rightarrow$ smallest

From correction video

For symm. matrix

If eig. values are distinct:

1, 2, 3

corresp. eig. vecs will be orthogonal (so obv. indep as well)

If repeated eig. vals.

1, 2, 2

eig. vecs. for diff. ones (1 & 2) will be ortho.

for same ones (2 and 2)

(A - 2I) ← we can choose 2 ortho. vecs. in null space of A - 2I

(this is for Symm. matrix)

For non-symm. matrix:

If distinct eig. vals: corresp. eig. vecs. indep.

NOT necessarily orthogonal.